

Lab Report No 1
Setting up Environment and getting started with
Python
Digital Signal Processing



Submitted By:

[Huzaifa Shahbaz]

[21-CE-009]

Submitted to:

Engr. Hareem Khan

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Department of Computer Engineering,
HITEC University, Taxila

Lab Task No A:

Solution:

Brief description (3-5 lines)
Install Anaconda Spyder and mentioned all steps in your lab report.
The explanation
1.Download the Anaconda installer. 2.Go to your Downloads folder and double-click the installer to launch. To prevent permission errors, do not launch the installer from the Favourites folder. 3.Click Next. 4.Read the licensing terms and click I Agree. 5.It is recommended that you install for Just Me, which will install Anaconda Distribution to just the current user account. Only select an install for All Users if you need to install for all users' accounts on the computer (which requires Windows Administrator privileges). 6.Click Next. 7.Select a destination folder to install Anaconda and click Next. Install Anaconda to a directory path that does not contain spaces or Unicode characters. 8.Choose whether to add Anaconda to your PATH environment variable or register Anaconda as your default Python. We don't recommend adding Anaconda to your PATH environment variable, since this can interfere with other software. Unless you plan on installing and running multiple versions of Anaconda or multiple versions of Python, accept the default and leave this box checked. Instead, use Anaconda software by opening Anaconda Navigator or the Anaconda Prompt from the Start Menu. 9.Click Install. If you want to watch the packages Anaconda is installing, click Show Details. 10.Click Next. 11.After a successful installation you will see the "Thanks for installing Anaconda" dialog box. 12.Click the Finish button.

Lab Task No B (1(A)):

Solution:

Brief description (3-5 lines)

Create a list and perform the following methods

- | |
|---|
| 1) insert()
2) remove()
3) append()
4) len()
5) pop()
6) clear() |
|---|

The code

<pre>a=[1,3,5,6,7,[3,4,5],"hello"] print(a) a.insert(3,20) print(a) a.remove(7) a.append("hi") print(a) a.pop() print(a) a.pop(6) print(a) a.clear() print(a)</pre>

The results (Screenshot)

<pre>[1, 3, 5, 6, 7, [3, 4, 5], 'hello'] [1, 3, 5, 20, 6, 7, [3, 4, 5], 'hello'] [1, 3, 5, 20, 6, [3, 4, 5], 'hello'] [1, 3, 5, 20, 6, [3, 4, 5], 'hello', 'hi'] [1, 3, 5, 20, 6, [3, 4, 5], 'hello', 'hi'] [1, 3, 5, 20, 6, [3, 4, 5], 'hello'] [1, 3, 5, 20, 6, [3, 4, 5]] []</pre>	
---	--

Lab Task No B(1(B)):

Solution:

Brief description (3-5 lines)

Create a dictionary and apply the following methods

- | |
|-------------------------------|
| 1) Print the dictionary items |
| 2) access items |
| 3) use get() |
| 4) change values |
| 5) use len() |

The code

<pre>1 my_dict={'huzaifa':2,'hamza':3,'apple':5} 2 print(my_dict) 3 print(my_dict['huzaifa']) 4 print(my_dict.get('orange',0)) 5 print(my_dict.get('apple',0)) 6 my_dict['huzaifa']=4 7 print(my_dict) 8 print(len(my_dict))</pre>
--

The results (Screenshot)

<pre>{'huzaifa': 2, 'hamza': 3, 'apple': 5} 2 0 5 {'huzaifa': 4, 'hamza': 3, 'apple': 5} 3</pre>
--

Lab Task No B(1(C)):

Solution:

Brief description (3-5 lines)

Create a tuple and perform the following methods

- 1) Add items
- 2) len()
- 3) check for item in tuple
- 4) Access items

The code

```
1 fruits=('apple','banana','cherry')
2 fruits+=('orange')
3 print(len(fruits))
4 print('banana' in fruits)
5 print('grape' in fruits)
6 print(fruits[0])
7 print(fruits[-1])
```

The results (Screenshot)

```
('apple', 'banana', 'cherry', 'orange')
4
True
False
apple
orange
```

Lab Task No B(2(A)):

Solution:

Brief description (3-5 lines)

Write a python program to add two numbers.

The code

```
1 a=int(input("enter the value for a"))
2 b=int(input("enter the value for a"))
3 c=a+b
4 print("The sum of a and b is",c)
```

The results (Screenshot)

<pre>enter the value for a:5 enter the value for a:10 The sum of a and b is 15</pre>
--

Lab Task No B(2(B)):

Solution:

Brief description (3-5 lines)

Write a python program to print a number is positive/negative using if-else.
--

The code

<pre>1 num=int(input("enter a number:")) 2 if num>0: 3 print("The number is positive:") 4 elif num==0: 5 print("The number is zero:") 6 else: 7 print("The number is negative:")</pre>

The results (Screenshot)

<pre>enter a number:8 The number is positive:</pre>

Lab Task No B(2(C)):

Solution:

Brief description (3-5 lines)

Write a python program to find largest number among three numbers.
--

The code

```

1 num1=int(input("enter the first number:"))
2 num2=int(input("enter the second number:"))
3 num3=int(input("enter the third number:"))
4 if num1>num2 and num1>num3:
5     largest>=num1
6 elif num2>num1 and num2>num3:
7     largest>=num2
8 else:
9     largest=num3
10 print("the largest number is:",largest)

```

The results (Screenshot)

```

enter the first number:55
enter the second number:44
enter the third number:88
the largest number is: 88

```

Lab Task No B(2(D)):

Solution:

Brief description (3-5 lines)
Write a python Program to read a number and display corresponding day using if_elif_else?
The code

```

1 day_num=int(input("enter a number 1 and 7:"))
2 if day_num==1:
3     day="Monday"
4 elif day_num==2:
5     day="Tuesday"
6 elif day_num==3:
7     day="Wednesday"
8 elif day_num==4:
9     day="Thursday"
10 elif day_num==5:
11     day="Friday"
12 elif day_num==6:
13     day="Saturday"
14 elif day_num==7:
15     day="Sunday"
16 else:
17     day="Invalid input"
18 print("The corresponding is:",day)

```

The results (Screenshot)

```

enter a number 1 and 7:6
The corresponding is: Saturday

```

Lab Task No B(3(A)):

Solution:

Brief description (3-5 lines)

Write a program to create a menu with the following options

1. TO PERFORM ADDITION
2. TO PERFORM SUBTRACTION
3. TO PERFORM MULTIPLICATION
4. TO PERFORM DIVISION

Accepts users input and perform the operation accordingly. Use functions with arguments.

The code

```
1  def add(a,b):
2      |         return a+b
3  def sub(c,d):
4      |         return c-d
5  def mul(e,f):
6      |         return e*f
7  def div(g,h):
8      |         return g/h
9  print("=====")
10 print("1.TO PERFORM ADDITION")
11 print("2.TO PERFORM SUBTRACTION")
12 print("3.FOR MULTIPLICATION")
13 print("4.FOR DIVISION")
14 choice=int(input("Enter your choice:"))
15 if choice==1:
16     |         a=int(input("Enter 1st value:"))
17     |         b=int(input("Enter 2nd value:"))
18     |         print(add(a,b))
19 elif choice==2:
20     |         c=int(input("Enter the 1st value:"))
21     |         d=int(input("Enter the 2nd value:"))
22     |         print(sub(c,d))
23 elif choice==3:
24     |         e=int(input("Enter the 1st value:"))
25     |         f=int(input("Enter the 2nd value:"))
26     |         print(mul(e,f))
27 elif choice==4:
28     |         g=int(input("enter the 1st value:"))
29     |         h=int(input("enter the 2nd value:"))
30     |         print(div(g,h))
31 else:
32     |         print("wrong choice")
```

The results (Screenshot)

```
=====
1.TO PERFORM ADDITION
2.TO PERFORM SUBTRACTION
3.FOR MULTIPLICATION
4.FOR DIVISION
Enter your choice:3
Enter the 1st value:55
Enter the 2nd value:66
3630
```

Lab Task No B(3(B)):

Solution:

Brief description (3-5 lines)

Write a python program to check whether the given string is palindrome or not.

The code

```
1 def is_palindrome(s):
2     s=s.replace(' ', '').lower()
3     return s==s[::-1]
4 s1="racecar"
5 s2="hello world"
6 print(f"{s1} is palindrome:{is_palindrome(s1)}")
7 print(f"{s2} is palindrome:{is_palindrome(s2)}")
```

The results (Screenshot)

```
racecar is palindrome:True
hello world is palindrome:False
```

Lab Task No B(3(C)):

Solution:

Brief description (3-5 lines)

Write a python program to find factorial of a given number using functions.

The code

```

1 def factorial(n):
2     if n == 0:
3         return 1
4     else:
5         return n*factorial(n-1)
6 num=int(input("Enter a number:"))
7 fact=factorial(num)
8 print ("The factorial of:", num, "is",fact)

```

The results (Screenshot)

```

Enter a number:6
The factorial of: 6 is 720

```

Lab Task No B(3(D)):

Solution:

Brief description (3-5 lines)

Write a Python function that takes two lists and returns True if they are equal otherwise false.

The code

```

1 def lists_are_equal(list1,list2):
2     if len(list1) != len(list2):
3         return False
4     for i in range(len(list1)):
5         if list1[i] != list2[i]:
6             return False
7     return True
8 list1=[1,2,3]
9 list2=[1,2,3]
10 if lists_are_equal(list1,list2):
11     print("The lists are equal")
12 else:
13     print("The lists are not equal")

```

The results (Screenshot)

The lists are equal

Lab Task No B(4(A)):

Solution:

Brief description (3-5 lines)
Write a program to double a given number and add two numbers using lambda()?
The code
<pre>1 double=lambda x:2*x 2 print(double(5)) 3 add=lambda x,y:x+y 4 print(add(5,4))</pre>
The results (Screenshot)

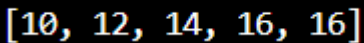
Lab Task No B(4(B)):

Solution:

Brief description (3-5 lines)
Write a program for filter() to filter only even numbers from a given list.
The code
<pre>1 numbers=[1,2,3,4,5,6,7,8,9,10] 2 even_numbers=list(filter(lambda x:x%2==0,numbers)) 3 print(even_numbers)</pre>
The results (Screenshot)

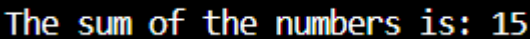
Lab Task No B(4(C)):

Solution:

Brief description (3-5 lines)
Write a program for map() function to double all the items in the list?
The code
<pre>1 my_list=[5,6,7,8,8] 2 doubled_list=list(map(lambda x:x*2,my_list)) 3 print(doubled_list)</pre>
The results (Screenshot)


Lab Task No B(4(D)):

Solution:

Brief description (3-5 lines)
Write a program to find sum of the numbers for the elements of the list by using reduce()?
The code
<pre>1 from functools import reduce 2 numbers=[1,2,3,4,5] 3 sum_of_numbers=reduce(lambda x,y:x+y,numbers) 4 print("The sum of the numbers is:",sum_of_numbers)</pre>
The results (Screenshot)


Lab Task No B(5(A)):

Solution:

Brief description (3-5 lines)
Demonstrate a python code to implement abnormal termination?
The code

```

1  a=5
2  b=0
3  print(a/b)
4  print("bye")

```

The results (Screenshot)

```

FILE 0:/ALL PROGRAMMING LANGUAGES
print(a/b)
ZeroDivisionError: division by zero

```

Lab Task No B(5(B)):

Solution:

Brief description (3-5 lines)

Demonstrate a python code to print try, except and finally block statements.

The code

```

1  try:
2  |      x=1/0
3  except ZeroDivisionError:
4  |      print("Cannot divide by zero!")
5  finally:
6  |      print("This code block is always executed.")

```

The results (Screenshot)

```

Cannot divide by zero!
This code block is always executed.

```

Lab Task No B(5(C)):

Solution:

Brief description (3-5 lines)

Write a python program to open and write “hello world” into a file?

The code

```
1 with open("hello.txt","w") as file:
2     file.write("hello world")
```

The results (Screenshot)

hello world

Lab Task No B(5(D)):

Solution:

Brief description (3-5 lines)

Write a python program to write the content “hi python programming” for the existing file.

The code

```
1 with open("filename.txt","w") as f:
2     f.write("hi python programming")
```

The results (Screenshot)

hi python programming

Lab Task No B(6(A)):

Solution:

Brief description (3-5 lines)

Write a python program to get python version.

The code

```

1 import sys
2 print("System version is:")
3 print(sys.version)
4 print("Version Information is:")
5 print(sys.version_info)

```

The results (Screenshot)

```

System version is:
3.6.2 (v3.6.2:5fd33b5, Jul 8 2017, 04:14:34) [MSC v.1900 32 bit (Intel)]
Version Information is:
sys.version_info(major=3, minor=6, micro=2, releaselevel='final', serial=0)

```

Lab Task No B(6(B)):

Solution:

Brief description (3-5 lines)

Write a python program to open a file and check what are the access permissions acquire by that file using os module?

The code

```

1 import os
2 filename="example.txt"
3 if not os.path.exists(filename):
4     file_perms=os.stat(filename).st_mode
5 if os.access(filename, os.R_OK):
6     file_perms|=0o400
7 if os.access(filename,os.W_OK):
8     file_perms|=0o200
9 if os.access(filename,os.X_OK):
10    file_perms|=0o100
11    print(f"File'{filename}'has permission:{oct(file_perms &0o777)}")
12 else:
13    print(f"File'{filename}'does not exist.")

```

The results (Screenshot)

```

File 'example.txt' does not exist.

```

Lab Task No B(6(C)):

Solution:

Brief description (3-5 lines)
Write a python program to display a particular month of a year using calendar module.
The code
<pre>1 import calendar 2 year=int(input("Enter the year:")) 3 month=int(input("Enter the month:")) 4 print(calendar.month(year,month))</pre>
The results (Screenshot)

Lab Task No B(6(D)):

Solution:

Brief description (3-5 lines)
Write a python program to print all the months of given year.
The code

```

1 import calendar
2 year=int(input("Enter the year: "))
3 for month in range(1,13):
4     print(calendar.month_name[month],year)

```

The results (Screenshot)

```

Enter the year: 2024
January 2024
February 2024
March 2024
April 2024
May 2024
June 2024
July 2024
August 2024
September 2024
October 2024
November 2024
December 2024

```

Lab Task No B(7(A)):

Solution:

Brief description (3-5 lines)

Write a python program to print date, time for today and now.

The code

```

1 import datetime
2 a=datetime.datetime.today()
3 b=datetime.datetime.now()
4 print(a)
5 print(b)

```

The results (Screenshot)

2024-02-25 16:16:50.376781 2024-02-25 16:16:50.376781	
--	--

Lab Task No B(7(B)):

Solution:

Brief description (3-5 lines)

Write a python program to add some days to your present date and print the date added.
--

The code

<pre>1 import datetime 2 days_to_add=5 3 current_date=datetime.date.today() 4 new_date=current_date+datetime.timedelta(days=days_to_add) 5 print("Current Date: ", current_date.strftime("%Y-%m-%d")) 6 print("New date:",new_date.strftime("%Y-%m-%d"))</pre>
--

The results (Screenshot)

Current Date: 2024-02-25 New date: 2024-03-01	
--	--

Lab Task No B(7(C)):

Solution:

Brief description (3-5 lines)

Write a python program to print date, time using date and time functions
--

The code

<pre>1 import datetime 2 now=datetime.datetime.now() 3 print("Current Date and Time is :", now.strftime("%Y-%m-%d %H:%M:%S"))</pre>

The results (Screenshot)

Current Date and Time is : 2024-02-25 16:26:33	
--	--

Lab Task No B(7(D)):

Solution:

Brief description (3-5 lines)
Write a python program which accepts the radius of a circle from user and computes the area (use math module).
The code
<pre>1 import math 2 radius=float(input("Enter the radius of the circle:")) 3 area=math.pi*radius**2 4 print(f"The area of the circle with radius{radius} is{area:.2f}")</pre>
The results (Screenshot)
<pre>Enter the radius of the circle:8 The area of the circle with radius8.0 is201.06</pre>

Lab Task No B(8(A)):

Solution:

Brief description (3-5 lines)
Write a python program to create a package (college),sub-package (alldept),modules(it,cse) and create admin and cabin function to module?
The code
<pre>1 def admin(): 2 print("hi") 3 def cabin(): 4 print("hello")</pre>
The results (Screenshot)
<pre>This is the admin function in the IT department. This is the cabin function in the CSE department.</pre>

Lab Task No B(8(B)):

Solution:

Brief description (3-5 lines)
Write a python program to create a package (Engg), sub-package (years), modules (sem) and create staff and student function to module?
The code
<pre>mkdir Engg mkdir Engg/years def staff(name): print("Staff name is:", name) def student(name): print("Student name is:", name) touch Engg/__init__.py from .sem import staff, student from Engg.years.sem import staff, student staff("REHAN") student("HUZAIFA")</pre>
The results (Screenshot)
<pre>Staff name is: REHAN Student name is: HUZAIFA</pre>

Lab Task No B(9(A)):

Solution:

Brief description (3-5 lines)
Write a python Program to display welcome to MRCET by using classes and objects.
The code
<pre>1 class display: 2 def displayMethod(self): 3 print("Welcome to mrcet") 4 obj=display() 5 obj.displayMethod()</pre>
The results (Screenshot)

Lab Task No B(9(B)):

Solution:

Brief description (3-5 lines)

Write a python Program to call data member and function using classes and objects.

The code

```
class Car:
    def __init__(self, make, model, year):
        self.make = make
        self.model = model
        self.year = year
        self.odometer_reading = 0
    def get_descriptive_name(self):
        long_name = f"{self.year} {self.make} {self.model}"
        return long_name.title()
    def read_odometer(self):
        print(f"This car has {self.odometer_reading} miles on it.")
    def update_odometer(self, mileage):
        if mileage >= self.odometer_reading:
            self.odometer_reading = mileage
        else:
            print("You can't roll back an odometer!")
    def increment_odometer(self, miles):
        self.odometer_reading += miles
my_car = Car("audi", "a4", 2020)
print(my_car.get_descriptive_name())
my_car.read_odometer()
my_car.update_odometer(1000)
my_car.read_odometer()
my_car.increment_odometer(500)
my_car.read_odometer()
```

The results (Screenshot)

```
2020 Audi A4  
This car has 0 miles on it.  
This car has 1500 miles on it.  
This car has 2500 miles on it.
```

Lab Task No B(9(C)):

Solution:

Brief description (3-5 lines)
Write a program to find sum of two numbers using class and methods
The code
<pre>class Calculator: def add(self, a, b): return a + b calculator = Calculator() num1 = int(input("Enter the first number: ")) num2 = int(input("Enter the second number: ")) sum = calculator.add(num1, num2) print("The sum of", num1, "and", num2, "is", sum)</pre>
The results (Screenshot)
<pre>Enter the first number: 55 Enter the second number: 88 The sum of 55 and 88 is 143</pre>

Lab Task No B(9(D)):

Solution:

Brief description (3-5 lines)
Write a program to read 3 subject marks and display pass or failed using class and object.
The code

```

class Student:
    def __init__(self):
        self.marks = []
    def add_marks(self):
        for i in range(3):
            self.marks.append(int(input(f"Enter marks for subject {i+1}: ")))
    def calculate_percentage(self):
        total_marks = sum(self.marks)
        percentage = (total_marks/300)*100
        return percentage
    def check_pass_fail(self):
        percentage = self.calculate_percentage()
        if percentage >= 40:
            print(f"Congratulations! You have passed with {percentage}%")
        else:
            print(f"Sorry, you have failed with {percentage}%")
student = Student()
student.add_marks()
student.check_pass_fail()

```

The results (Screenshot)

```

Enter marks for subject 1: 55
Enter marks for subject 2: 88
Enter marks for subject 3: 98
Congratulations! You have passed with 80.33333333333333%

```

Lab Task No B(10(A)):

Solution:

Brief description (3-5 lines)

Using a numpy module create an array and check the following:

1. Type of array
2. Axes of array
3. Shape of array
4. Type of elements in array

The code


```
import numpy as np
arr=np.array([[1,2,3],[4,2,5]])
print("Array is of type:",type(arr))
print("no.of dimensions:",arr.ndim)
print("Shape of array:",arr.shape)
print("Size of array:",arr.size)
print("Array stores elements of type:",arr.dtype)
```

The results (Screenshot)

```
Array is of type: <class 'numpy.ndarray'>
no.of dimensions: 2
Shape of array: (2, 3)
Size of array: 6
Array stores elements of type: int32
```

Lab Task No B(10(B)):

Solution:

Brief description (3-5 lines)
Using a numpy module create array and check the following:
1. List with type float
2. 3*4 array with all zeros
3. From tuple
4. Random values
The code

```
import numpy as np
lst = [1, 2, 3, 4, 5]
arr = np.array(lst, dtype=float)
print(arr)
print(arr.dtype)
import numpy as np
arr = np.zeros((3, 4))
print(arr)
import numpy as np
tup = (1, 2, 3, 4, 5)
arr = np.array(tup)
print(arr)
import numpy as np
arr = np.random.rand(3, 4)
print(arr)
```

The results (Screenshot)

```
[1. 2. 3. 4. 5.]
float64
[[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]
[1 2 3 4 5]
[[0.02849096 0.8273807 0.18994557 0.9055901 ]
 [0.45567775 0.94857713 0.59172913 0.31178535]
 [0.78684173 0.50163506 0.5856282 0.31456791]]
```

Lab Task No B(10(C)):

Solution:

Brief description (3-5 lines)

Using a numpy module create array and check the following:

1. Reshape 3X4 array to 2X2X3 array
2. Sequence of integers from 0 to 30 with steps of 5
3. Flatten array

4. Constant value array of complex type

The code

```
import numpy as np
arr1 = np.array([[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12]])
arr2 = arr1.reshape(2, 2, 3)
print("Reshaped array:")
print(arr2)
print()
arr3 = np.arange(0, 31, 5)
print("Sequence of integers:")
print(arr3)
print()
arr4 = arr2.flatten() |
print("Flattened array:")
print(arr4)
print()
arr5 = np.full((3, 3), 2+3j, dtype=complex)
print("Constant value array of complex type:")
print(arr5)
```

The results (Screenshot)

```
Reshaped array:
[[[ 1  2  3]
  [ 4  5  6]]

 [[ 7  8  9]
  [10 11 12]]]

Sequence of integers:
[ 0  5 10 15 20 25 30]

Flattened array:
[ 1  2  3  4  5  6  7  8  9 10 11 12]

Constant value array of complex type:
[[2.+3.j 2.+3.j 2.+3.j]
 [2.+3.j 2.+3.j 2.+3.j]
 [2.+3.j 2.+3.j 2.+3.j]]
```

Lab Task No B(11(A)):

Solution:

Brief description (3-5 lines)
Write a python program to concatenate the data frames with two different objects.
The code
<pre>import pandas as pd one=pd.DataFrame({'Name':['teju','gouri'], 'age':[19,20]}, index=[1,2]) two=pd.DataFrame({'Name':['suma','nammu'], 'age':[20,21]}, index=[3,4]) print(pd.concat([one,two]))</pre>
The results (Screenshot)

	Name	age
1	teju	19
2	gouri	20
3	suma	20
4	nammu	21

Lab Task No B(11(B)):

Solution:

Brief description (3-5 lines)

Write a python code to read a csv file using pandas module and print the first and last five lines of a file.

The code

```
import pandas as pd
df = pd.read_csv('filename.csv')
print('First Five Rows:\n', df.head())
print('Last Five Rows:\n', df.tail())
```

The results (Screenshot)

```
df = pd.read_csv('filename.csv')

File "E:\Python\PYTHON (2)\lib\site-packages\pandas\util\_decorators.py", line 311,
    return func(*args, **kwargs)

File "E:\Python\PYTHON (2)\lib\site-packages\pandas\io\parsers\readers.py", line 67,
    return _read(filepath_or_buffer, kwds)

File "E:\Python\PYTHON (2)\lib\site-packages\pandas\io\parsers\readers.py", line 57,
    parser = TextFileReader(filepath_or_buffer, **kwargs)

File "E:\Python\PYTHON (2)\lib\site-packages\pandas\io\parsers\readers.py", line 93,
    self._engine = self._make_engine(f, self.engine)

File "E:\Python\PYTHON (2)\lib\site-packages\pandas\io\parsers\readers.py", line 12,
make_engine
    self.handles = get_handle( # type: ignore[call-overload]
```

Lab Task No B(12(A)):

Solution:**Brief description (3-5 lines)**

Write a python code to set background color and pic and draw a circle using turtle module.

The code

```
import turtle  
t=turtle.Turtle  
t.circle(50)  
s=turtle.Screen  
s.bgcolor("black")  
s.bgpic("pic.gif") |
```

The results (Screenshot)

Lab Task No B(12(B)):

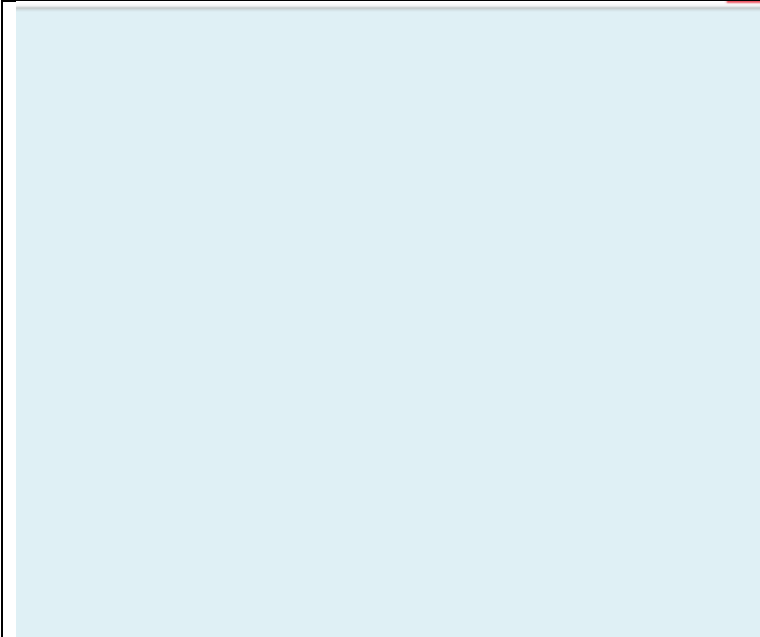
Solution:**Brief description (3-5 lines)**

Write a python code to set background color and pic and draw a square and fill the color using turtle module.

The code

```
import turtle
wn = turtle.Screen()
wn.bgcolor("lightblue")
wn.bgpic("background.gif")
# Set up the turtle
t = turtle.Turtle()
t.shape("turtle")
t.color("red")
for i in range(4):
    t.forward(100)
    t.right(90)
t.begin_fill()
for i in range(4):
    t.forward(100)
    t.right(90)
t.end_fill()
turtle.done()
```

The results (Screenshot)



Lab Task No B(12(C)):

Solution:

Brief description (3-5 lines)

Write a python code to perform addition using functions with pdb module.

The code

```
import pdb
def add_numbers(a, b):
    pdb.set_trace()
    result = a + b
    return result
x = 5
y = 10
sum = add_numbers(x, y)
print("The sum of", x, "and", y, "is", sum)
```

The results (Screenshot)

```
In [1]: runfile('E:/Python/import pdb.py', wdir='E:/Python')
> e:\python\import pdb.py(6)add_numbers()
4 def add_numbers(a, b):
5     pdb.set_trace()
----> 6     result = a + b
7     return result
8 x = 5

IPdb [1]: 5
5

IPdb [2]: 2
2

IPdb [3]: 6
```

Conclusion

Setting up an environment and getting started with Python is a relatively straightforward process. With a little practice, you will soon be able to write programs that can solve real-world problems.